

ARCHITECTURE & STRATEGY

MIDDLEBOX DEPLOYMENT

- **NGINX Proxy Debian VM:** Positioned strategically in the DMZ between Router R1 and Router R3.
- **Network Segregation:** Intercepts all traffic, breaking direct external-to-internal IP connections.
- **DMZ Isolation:** Keeps the Internal Web Servers strictly invisible to the Internet.

SECURITY PRINCIPLES

Zero Trust

Applies the "**Least Privilege**" rule. The external client (**Malicious PC**) can only interact with the **Debian Proxy**, never the backend.

WAF: HARDENING (PHASE 1)

ENCRYPTION & INTEGRITY

- **SSL/TLS Implementation:** The **Debian Proxy** acts as the termination point, securing data in transit from the **Malicious PC**.
- **Confidentiality & Integrity:** Encrypts payloads and uses hashing to ensure data isn't tampered with over the public network.

TRAFFIC CONTROL (DOS)

- **Rate Limiting:** The **Debian Proxy** restricts requests per external IP to prevent resource exhaustion.
- **Payload Size Limits:** Blocks large request bodies to protect **Internal Web Servers** from buffer overflow attempts.

WAF: HARDENING (PHASE 2)

INFORMATION HIDING

- **Masking Version:** The **Debian Proxy** hides its NGINX version string to block CVE-based reconnaissance.
- **Anti-Footprinting:** Increases difficulty for the **Malicious PC** to identify specific software vulnerabilities.

SECURE HEADERS

- **HSTS:** The **Debian Proxy** forces HTTPS to prevent Man-in-the-Middle (MitM) attacks.
- **XSS Protection:** Blocks unauthorized client-side script execution.
- **Conclusion:** We successfully isolated the **Internal Servers** behind a resilient perimeter.